Vierendeel girder

Plane truss — analysis and CIRSOC 301 rod checks

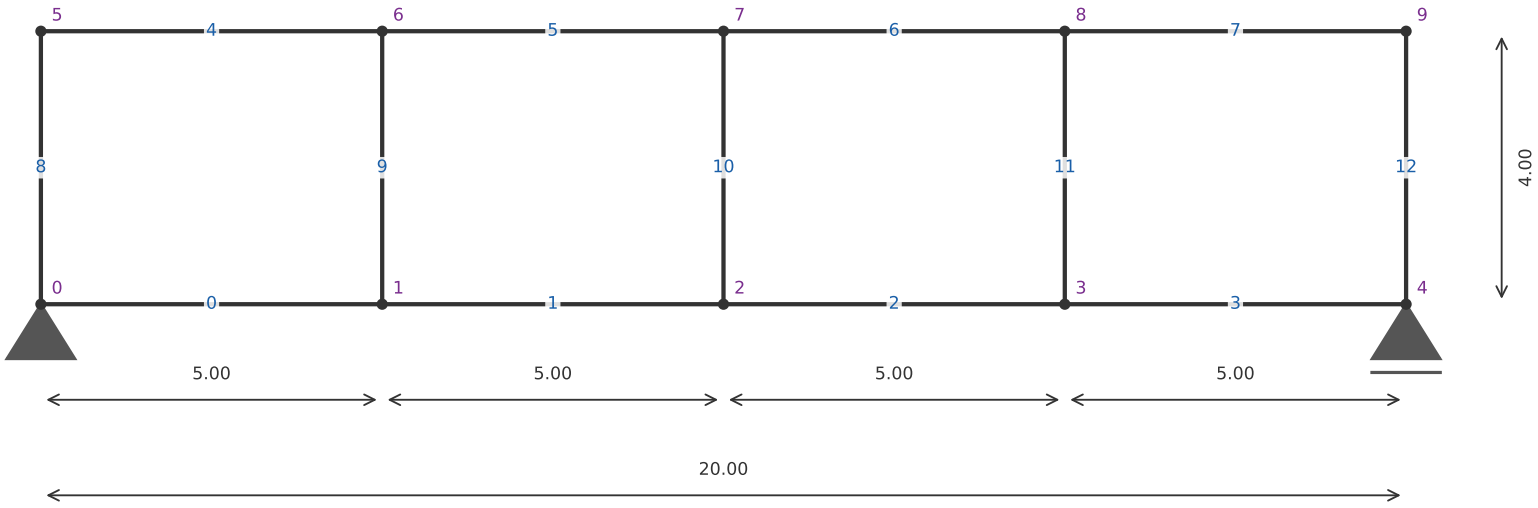
Span	20.00 m
Height	4.00 m
Nodes / rods	10 / 13
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 60.0 kN
Family "Frame"	SHS 200x200x8 (F24 (CIRSOC) / A36)
Weight	Weighs 2951 kg, carries 60.0 kN (2 × its own weight), passes (0.55).
Deflection	55.86 mm = L/358, limit L/300: OK
Stability	Statically indeterminate to degree 12 (deformations + r − DOF = 39 + 3 − 30 = 12): the forces also depend on the stiffness of the rods.
Most used rod	rod 9 at 0.55 (H1-1b, axial + bending)
Rods over capacity	0 of 13 checked
Buckling	$\lambda_1 = 144.35$
Shear and moment	present — see the Shear, Moment and Rod detail sheets

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4. Deflection and stress
5. Reactions
6. Utilisation
7. Buckling
8. Shear V
9. Moment M
10. Rods
11. Rod detail
12. Explain rod 9 and 11

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.



5 m · true to scale

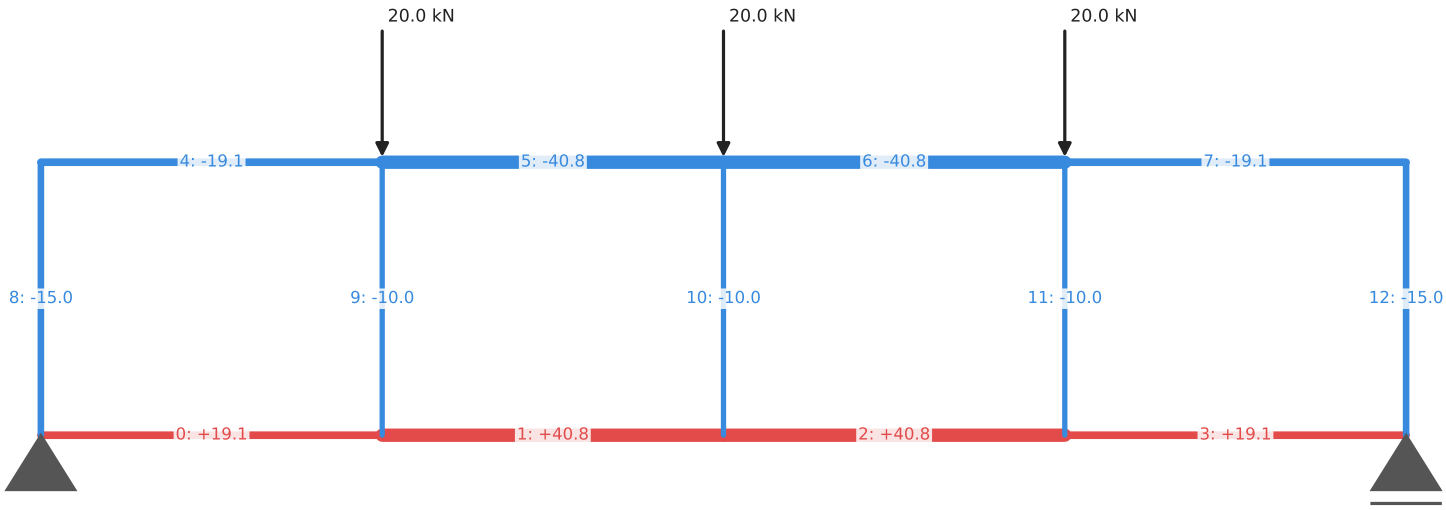
0 5

Truss — Geometry

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MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 40.8).



5 m · true to scale

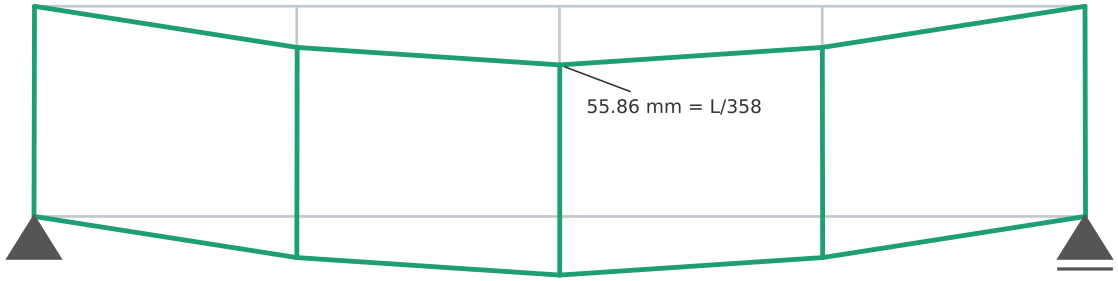
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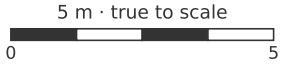
Truss — Axial forces

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MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

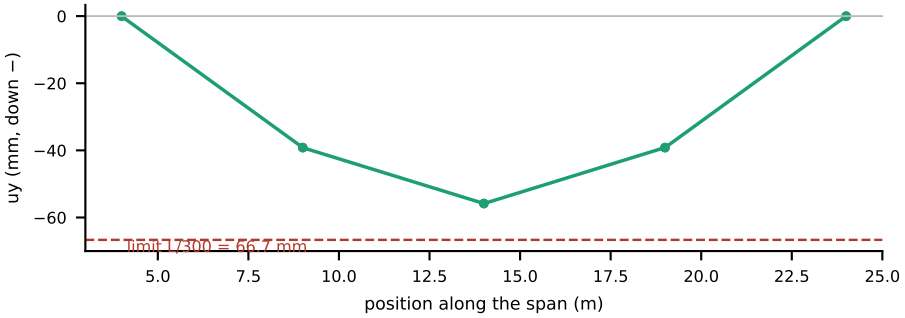
Deflected shape (green) over the truss (grey). Geometry to scale; displacements magnified × 20. Largest vertical deflection 55.86 mm against the limit L/300 = 66.7 mm: OK.





Elastic deformation

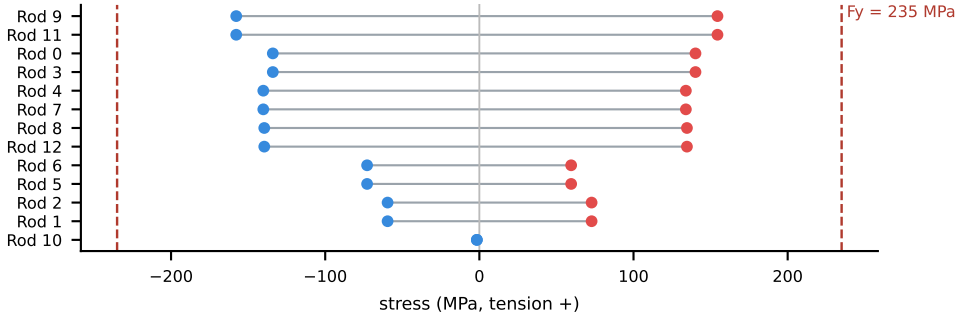
vertical movement of every joint



Largest 55.86 mm at joint 7 = L/358. Within the limit.

Fibre stress

N/A ± M·c/I at each rod's extreme fibres



The 13 most stressed rods; the bar between a rod's two dots is its bending.

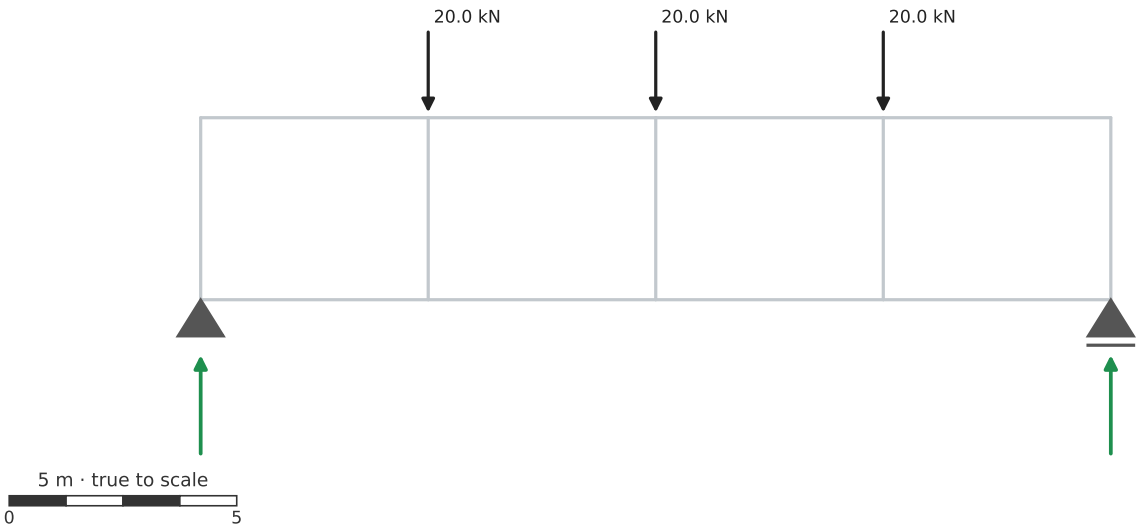
Truss — Deflection and stress

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MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Support reactions (green arrows), as the supports push on the truss; Ry positive upwards. The sums balance the loads.

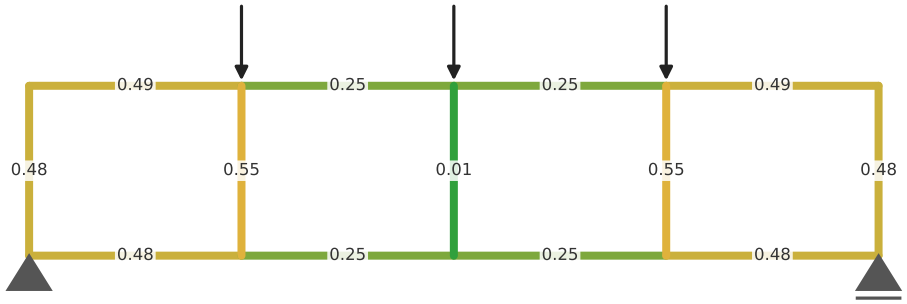
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	30.00
4	rollerX	0.00	30.00
Σ		0.00	60.00



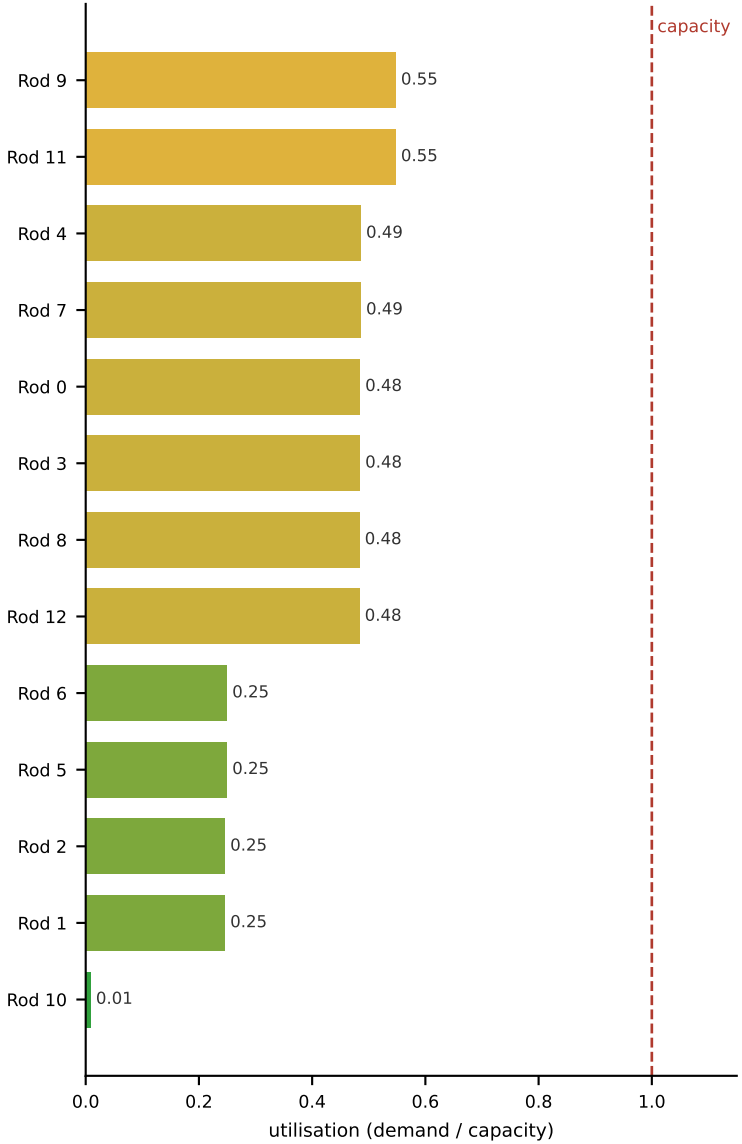
Truss — Reactions

MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Utilisation: demand over CIRSOC 301 capacity. Most used rod 9 at 0.55 (H1-1b, axial + bending). Every checked rod is within capacity.



5 m · true to scale
0 5





grey = no steel section, not checked

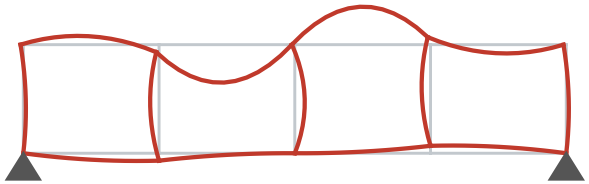
Truss — Utilisation

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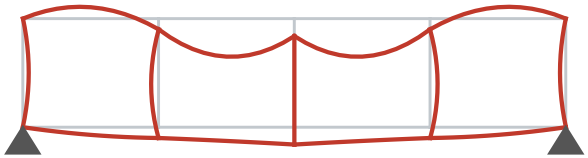
MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

λ is the factor on the load as it stands. λ_1 = 144.35, so the load can grow 144.3× before it buckles. Mode 2 is 209.39, 45 % higher: one member decides the buckling load.

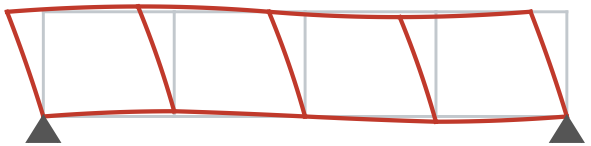
Mode 1: λ = 144.35
rod 5 bows

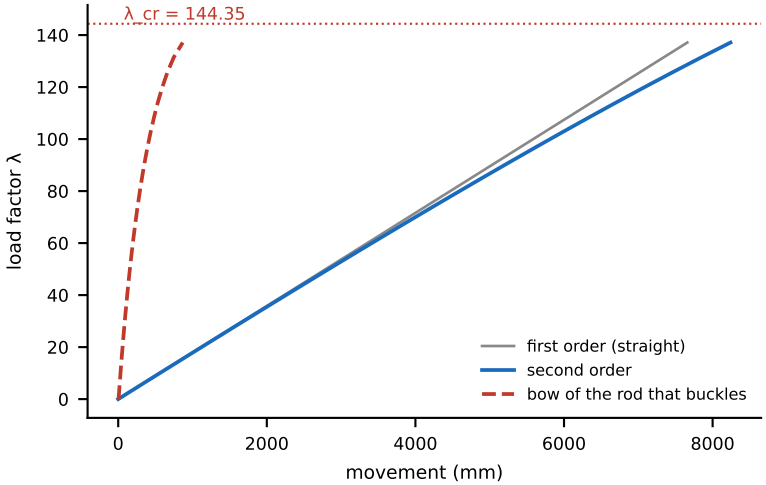


Mode 2: λ = 209.39
rod 5 bows



Mode 3: λ = 222.54
node 6 moves most





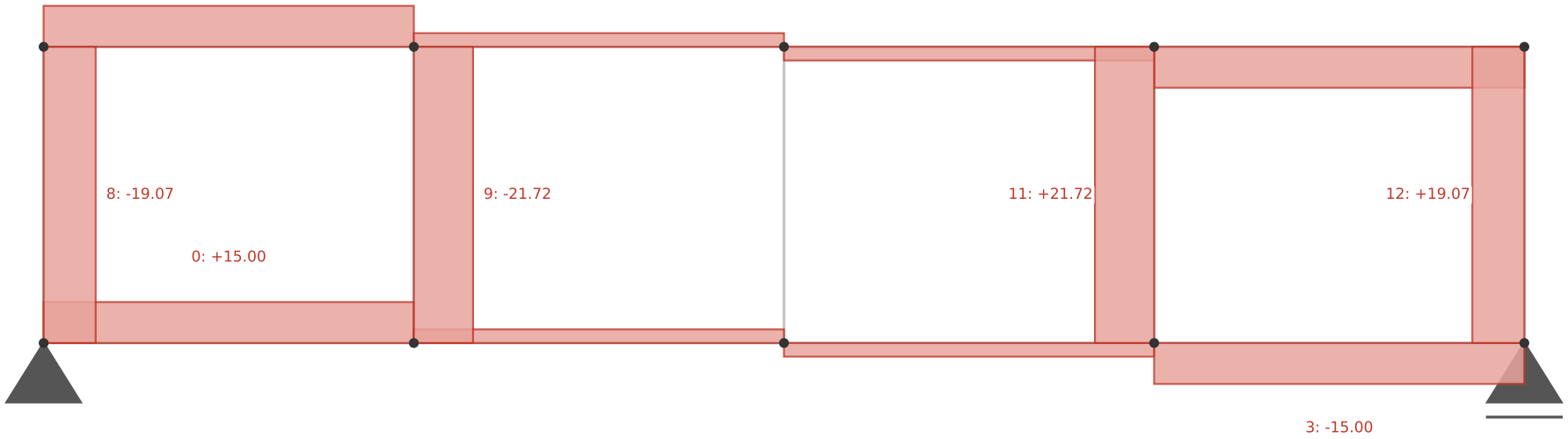
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.00 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Truss — Buckling

MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Shear force V along every rigid rod. Biggest +21.72 kN in rod 11; the band is drawn off each member, proportional to the value, and labelled on the six largest.



5 m · true to scale

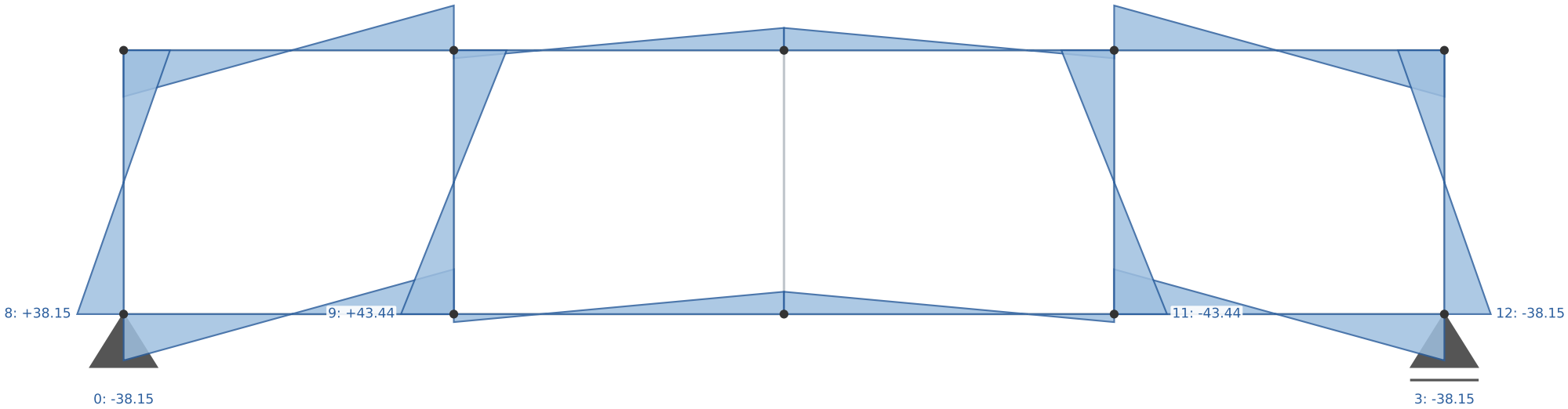
0 5

Truss — Shear V

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MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Bending moment M along every rigid rod (sagging positive). Biggest -43.44 kN·m in rod 11; the band is drawn off each member, proportional to the value, and labelled on the six largest.



5 m · true to scale

0 5

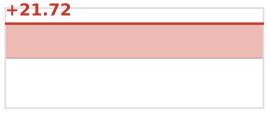
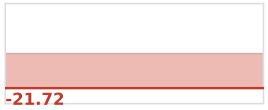
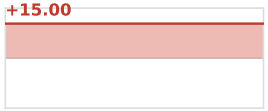
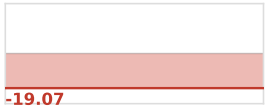
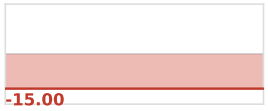
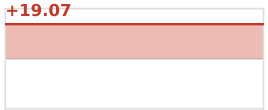
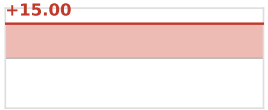
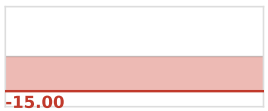
Truss — Moment M

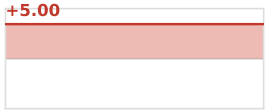
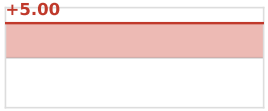
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MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use	
0	0–1	Frame	SHS 200x200x8	5.00	+19.07	H1-1b, axial + bending	0.48	OK
1	1–2	Frame	SHS 200x200x8	5.00	+40.79	H1-1b, axial + bending	0.25	OK
2	2–3	Frame	SHS 200x200x8	5.00	+40.79	H1-1b, axial + bending	0.25	OK
3	3–4	Frame	SHS 200x200x8	5.00	+19.07	H1-1b, axial + bending	0.48	OK
4	5–6	Frame	SHS 200x200x8	5.00	-19.07	H1-1b, axial + bending	0.49	OK
5	6–7	Frame	SHS 200x200x8	5.00	-40.79	H1-1b, axial + bending	0.25	OK
6	7–8	Frame	SHS 200x200x8	5.00	-40.79	H1-1b, axial + bending	0.25	OK
7	8–9	Frame	SHS 200x200x8	5.00	-19.07	H1-1b, axial + bending	0.49	OK
8	0–5	Frame	SHS 200x200x8	4.00	-15.00	H1-1b, axial + bending	0.48	OK
9	1–6	Frame	SHS 200x200x8	4.00	-10.00	H1-1b, axial + bending	0.55	OK
10	2–7	Frame	SHS 200x200x8	4.00	-10.00	inelastic buckling (E3-2)	0.01	OK
11	3–8	Frame	SHS 200x200x8	4.00	-10.00	H1-1b, axial + bending	0.55	OK
12	4–9	Frame	SHS 200x200x8	4.00	-15.00	H1-1b, axial + bending	0.48	OK

One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 11 use 0.55 3–8 · 4.00 m SHS 200x200x8 N -10.00 kN			
Rod 9 use 0.55 1–6 · 4.00 m SHS 200x200x8 N -10.00 kN			
Rod 0 use 0.48 0–1 · 5.00 m SHS 200x200x8 N +19.07 kN			
Rod 8 use 0.48 0–5 · 4.00 m SHS 200x200x8 N -15.00 kN			
Rod 3 use 0.48 3–4 · 5.00 m SHS 200x200x8 N +19.07 kN			
Rod 12 use 0.48 4–9 · 4.00 m SHS 200x200x8 N -15.00 kN			
Rod 4 use 0.49 5–6 · 5.00 m SHS 200x200x8 N -19.07 kN			
Rod 7 use 0.49 8–9 · 5.00 m SHS 200x200x8 N -19.07 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 6 use 0.25 7–8 · 5.00 m SHS 200x200x8 N -40.79 kN			
Rod 5 use 0.25 6–7 · 5.00 m SHS 200x200x8 N -40.79 kN			
Rod 2 use 0.25 2–3 · 5.00 m SHS 200x200x8 N +40.79 kN			
Rod 1 use 0.25 1–2 · 5.00 m SHS 200x200x8 N +40.79 kN			

Truss — Rod detail

MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58



Rod 9

utilisation 0.55

H1-1b, axial + bending

```
Rod 9
=====
Steel, CIRSOC 301-2018 -- A = 6,144 mm², Fy = 235 MPa, L = 4.000 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,000 mm / 78.5 mm
      = 51.0

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 51.0²
      = 759.30 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 51.0 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/759.30) × 235
      = 206.45 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 206.45 × 6,144
      = 1,141.57 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 10.00 / 1,141.57
      = 0.009

 7. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 378,143 mm³
      = 79.977 kN·m

 8. Axial + bending  [H1-1b]
      Pr/(2Pc) + (Mry/Mc + Mrx/Mcw)
      = 0.009/2 + (0.000/79.977 + 0.000/79.977)
      = 0.548

 9. Governs
      the largest ratio
      = H1-1b, axial + bending
      = 0.548
... 2 more lines; the verdict is in the heading.
```



Rod 11

utilisation 0.55

H1-1b, axial + bending

```
Rod 11
======
Steel, CIRSOC 301-2018 -- A = 6,144 mm², Fy = 235 MPa, L = 4.000 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,000 mm / 78.5 mm
      = 51.0

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 51.0²
      = 759.30 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 51.0 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/759.30) × 235
      = 206.45 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 206.45 × 6,144
      = 1,141.57 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 10.00 / 1,141.57
      = 0.009

 7. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 378,143 mm³
      = 79.977 kN·m

 8. Axial + bending  [H1-1b]
      Pr/(2Pc) + (Mry/Mc + Mrx/Mcw)
      = 0.009/2 + (0.000/79.977 + 0.000/79.977)
      = 0.548

 9. Governs
      the largest ratio
      = H1-1b, axial + bending
      = 0.548
... 2 more lines; the verdict is in the heading.
```

Truss — Explain rod 9 and 11

MODEL	SIZE	UNITS	CODE	DATE
Vierendeel girder	10 nodes / 13 rods	m, kN, mm	CIRSOC 301	2026-10-12 01:58